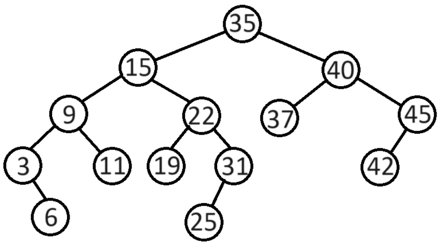
NAME: Algorithms and Data Structures 2.
Neptun: Sample questions for exam 2.

1 Type A) Build an AVL-tree with the following key values by inserting them one-by-one in the given order (starting from the empty tree). Show every insertion. If rebalancing is needed, then show the tree before the rebalancing (with the balance values of the nodes), name the type of rebalancing rotation that is used, and finally show the rebalanced tree with the balance values of the nodes. Give the parenthesized form of the final tree!

20 15 25 32 21 29 40 18 7 11 5 19

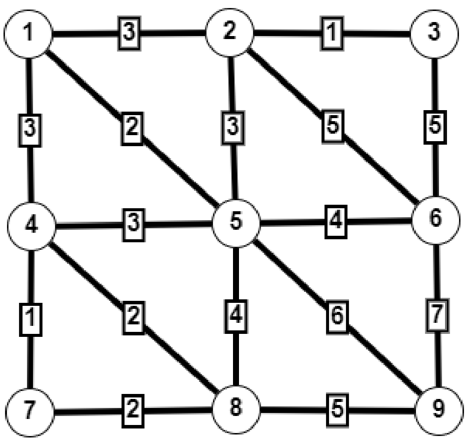
1 Type B) Write the balance values beside the nodes on the following tree. Is it an AVL tree? Show the steps of procedure AVLdel(40). If rebalancing is needed, then draw the tree before the rebalancing (with the balance values), show which subtree needs rebalancing, write the name of the rebalancing rotation that is used, and finally draw the rebalanced tree (with the balance values). If more than one rebalancing rotation is necessary, then do this for all of them. Give the parenthesized form of the final tree.



2) Illustrate Prim's algorithm on the following (undirected) graph. First draw the graph, then fill out a table as we did in practice, and finally draw the minimum spanning tree. (If you have a choice during the algorithm, then always select the vertex with smaller index.)

1	-	2, 2; 5, 1.	5	-	1, 1; 6, 1..
2	-	1, 2; 6, 0.	6	-	2, 0; 3, 1; 5, 1; 7, 2.
3	-	4, 4; 6, 1; 7, 1.	7	-	3, 1; 4, 3; 6, 2; 8, 1.
4	-	3, 4; 7, 3; 8, 2.	8	-	4, 2; 7, 1.

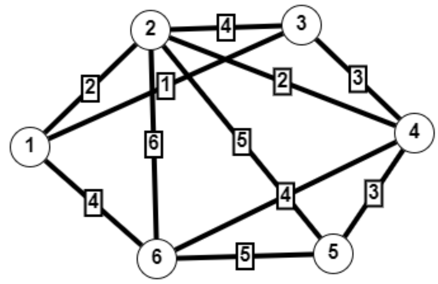
3) Find a minimal spanning tree in the following undirected weighted graph using Kruskal's algorithm. Complete the table. (In the "components" column list the components separated by commas.)



Components	edge	its weight	is it safe?

List the edges that are in the minimal spanning tree! What is the weight of this minimal spanning tree?

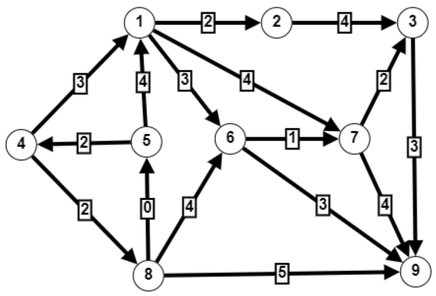
4) a) Present the adjacency matrix representation of the following graph:



b) Run Dijkstra's algorithm on this graph with vertex 1 as source. Illustrate the algorithm by completing a table like we did in practice. When you can choose between vertices, always choose the one with samller index.

c) Draw the shortest path's tree you obtained in b).

5) Let $G = (V, E)$ be the following graph:



a) Present the adjacency list representation of G.

b) Run procedure *topologicalSort*$(G, 1)$ on this graph (the modified DFS algorithm that produces a topological sort of the vertices that can be reached from vertex 1). Fill the table below with the $d(u)$ and $f(u)$ values after running the procedure.

	1	2	3	4	5	6	7	8	9
π									
d									
f									

c) Present the shortest paths algorithm for DAGs on this graph using 1 as source vertex. Fill a table as we did in practice.

6) Run the QBF algorithm on the graph given by adjacency matrix A. Select the first vertex as source.

$$A = \begin{pmatrix} 0 & 6 & \infty & 7 & \infty \\ \infty & 0 & 5 & 8 & -4 \\ \infty & -2 & 0 & \infty & \infty \\ \infty & \infty & -3 & 0 & 9 \\ \infty & \infty & 7 & \infty & 0 \end{pmatrix}$$

a) Draw a graphical representation of the graph.

b) Complete the following table.

changes of *d*, *e*					exp. vertex	Q:	changes of π				
1	2	3	4	5	*u* : *d(u)* : *e(u)*	*queue*	1	2	3	4	5
					--	--					

7) Illustrate the Floyd-Warshall algorithm on the graph given by adjacency matrix A.

$$A = \begin{pmatrix} 0 & 5 & 3 & 6 \\ 4 & 0 & \infty & \infty \\ \infty & -1 & 0 & -2 \\ 2 & -2 & 5 & 0 \end{pmatrix}$$

For each $k = 1..4$ present matrices D and π.

8) Design an algorithm which solves the following problem: you are given two directed weighted graphs G_1 and G_2 both having vertex set $V = \{1, 2, 3, \dots, n\}$. The graphs are given with adjacency list representation $A_1, A_2/1 : Edges^*[n]$ and their weight functions are w_1 and w_2 respectively. Your task is to produce the adjacency list representation of the graph $G_1 + G_2$, where

- the vertex set of $G_1 + G_2$ is $V = \{1, 2, 3, \dots, n\}$, and
- (u, v) is an edge of $G_1 + G_2$ if and only if it is an edge of G_1 or G_2 or both, and
- the weight of (u, v) is

  $w_1(u, v) + w_2(u, v)$ if (u, v) is an edge in both G_1 and G_2

  $w_1(u, v)$ if (u, v) is an edge in G_1 but not in G_2

  $w_2(u, v)$ if (u, v) is an edge in G_2 but not in G_1.

What is the time complexity of your algorithm? You can have full points for an algorithm with $\Theta(n^2)$ time complexity, or extra points for $\Theta(n + m)$.

Practical information about Exam 2.

- There will be 1 question about AVL trees: demonstration of insertion and/or deletion.
- 3 questions will be about demonstrating algorithms on weighted graphs given with different representations.
- There will be one question about designing an algorithm to solve a given problem.

Rules

- No mobile phones, no smart watches, no tablets allowed during the exam. You will have to put all these devices into your bag.
- You cannot use printed or written notes.
- No talking allowed to fellow students. If you need anything, or if you have a question, you can ask me, but not your fellow student.
- No bathroom brakes.
- The retake will take place on the week starting with the 16th of December. The exact date will be given by next week.
- If you need consultation, please write to the demonstrators in Teams.